STUDENT-FACING RESOURCE

LLM Resume Verification Framework

A human-controlled method for checking extracted resume entities without inventing qualifications

> **Core rule:** The LLM may flag evidence, omissions, and possible errors. The student must compare every result with the original resume and make the final decision.

Where this framework fits

1 | Prepare	2 | Extract	3 | Verify	4 | Confirm
Export the final resume as a text-based PDF.	Copy or extract the PDF text and preserve reading order notes.	Use the structured prompt to check system predictions.	Compare each result with the original PDF before revising.

Fixed entity labels

Use only these five labels. The full 1,020-resume recovery experiment focused on SKILL, while this student framework retains the complete project schema.

Label	Meaning	Resume example
PERSON	Applicant name	Jordan Lee
ORG	Employer, school, or organization	Bright Labs; Caldwell University
DATE	Employment or education date	Jun 2025-Aug 2025
GPE	City, state, or country	Newark, NJ; Nepal
SKILL	Explicit technical qualification	Python; Flask; PostgreSQL

Evidence rules

- **Exact evidence:** Keep a prediction only when its complete meaning is supported by the resume.
- **No inference:** Do not turn a responsibility, course title, or broad category into a skill that is never stated.
- **Truth before matching:** Never add a job-description keyword unless it accurately describes the student's experience.
- **Human authority:** Treat the model output as a review worksheet, not as a correct answer or an ATS guarantee.

PROMPT TEMPLATE

Prepare the evidence before asking the model

1. **De-identify when possible.** Replace the student's name, email, phone number, street address, and other unnecessary personal details with neutral placeholders.
2. **Create the text view.** Copy the PDF text into a plain-text editor. Keep the original wording even when the reading order is imperfect.
3. **Record visual problems.** Write short notes about columns, icons, image-only text, missing lines, or reordered sections that the plain text does not show.
4. **Add candidate predictions.** Paste the entities produced by the parser. Include the predicted text and entity type. If there are no predictions, write none.

> **Privacy:** Do not upload a real resume to an external AI service unless the student has permission and the institution's privacy rules allow it. Redact personal information whenever it is not needed for the check.

Copy-and-use verification prompt

```
You are a resume extraction verification assistant.

Use only the supplied resume text, visual notes, and system predictions. Do not infer, invent, recommend, or rewrite qualifications. A skill is valid only when the resume states it explicitly. Preserve the resume's wording whenever possible.

Allowed entity types: PERSON, ORG, DATE, GPE, SKILL.

For every system prediction, assign one status:
- verified_exact: the complete prediction appears in the supplied evidence.
- verified_partial: only part of the prediction is supported or its boundary is incomplete.
- unsupported: the prediction is not supported by the supplied evidence.

Also list clearly stated allowed entities that the system missed. Return valid JSON only, following the required output structure. If the evidence is insufficient, say so in the note instead of guessing.

RESUME_ID: [de-identified ID]
EXTRACTED_TEXT:
[paste the plain-text extraction]

VISUAL_NOTES:
[optional human notes about columns, images, icons, or reading order]

SYSTEM_PREDICTIONS:
[paste candidate entity text and type, or write none]
```

Research note: GPT-4o-mini was used for the full-corpus SKILL recovery phase. Record the exact model and date whenever this framework is used because model behavior can change.

EXPECTED OUTPUT

Require structured JSON, then read it as evidence

JSON makes each decision visible and reviewable. A valid response should separate supported predictions, missed entities, and document-level warnings.


```
{
  "resume_id": "STUDENT_SAMPLE",
  "verified_predictions": [
    {
      "text": "Bright Labs",
      "type": "ORG",
      "status": "verified_exact",
      "evidence": "Software Intern - Bright Labs",
      "section": "Experience",
      "note": ""
    }
  ],
  "missed_entities": [
    {
      "text": "PostgreSQL",
      "type": "SKILL",
      "evidence": "Built a Flask API with PostgreSQL",
      "section": "Projects"
    }
  ],
  "document_warnings": [
    "The extracted text places the Skills section inside an Experience bullet."
  ]
}
```


How to interpret each status

Status	Meaning	Student action
verified_exact	The full entity is directly supported.	Confirm it in the PDF and keep it.
verified_partial	Only part of the text or boundary is supported.	Correct the entity using the PDF wording.
unsupported	The supplied evidence does not support it.	Remove it from the verified set.
missed entity	The resume states an allowed entity that the system omitted.	Confirm it manually, then record the miss.
document warning	The extraction may have lost or reordered information.	Inspect the PDF and plain-text view together.

Do not use model confidence as proof: A confident-sounding response can still be wrong. This framework uses evidence and status labels instead of treating a self-reported confidence score as accuracy.

WORKED EXAMPLE

What verification looks like on a resume

Resume excerpt

> Software Intern - Bright Labs | Newark, NJ | Jun 2025-Aug 2025
> Built a Flask API in Python and PostgreSQL to manage 500+ records.

Candidate review

Candidate	Type	Decision	Why
Bright Labs	ORG	verified_exact	The organization appears exactly in the Experience entry.
Newark, NJ	GPE	verified_exact	The location appears directly beside the role.
Jun 2025-Aug 2025	DATE	verified_exact	The complete date range is present.
Flask	SKILL	verified_exact	The project bullet states the framework explicitly.
Python	SKILL	verified_exact	The project bullet states the language explicitly.
Java	SKILL	unsupported	Java does not appear in the supplied excerpt.

Manual verification protocol

1. **Open both views.** Keep the original PDF and plain-text extraction visible at the same time.
2. **Check every prediction.** Find the exact evidence in the resume and verify both the text boundary and entity type.
3. **Review every proposed miss.** Confirm that the term is explicitly stated and belongs to one of the five allowed entity types.
4. **Resolve disagreement manually.** When the LLM and parser disagree, use the original resume as the source of truth.
5. **Record the decision.** Save the model, date, prompt version, evidence, and final human status so the review can be repeated.

> **Correct conclusion:** The example supports Bright Labs, Newark, the date range, Flask, Python, and PostgreSQL. It does not support Java, even if Java appears in the target job description.

LIMITATIONS AND RECORD

What this framework cannot guarantee

- **Hallucination:** The LLM may invent evidence, accept unsupported terms, or omit information that is present.
- **Extraction failure:** The model cannot recover visual information that was never supplied in the text or visual notes.
- **Boundary changes:** The model may shorten, expand, or normalize a phrase instead of preserving the resume's exact wording.
- **Model variation:** Outputs can change across models, versions, prompts, and repeated runs.
- **ATS uncertainty:** Passing this check does not guarantee how a proprietary applicant tracking system or Jobscan will score the resume.
- **Research scope:** This is a practical verification method derived from the project; it is not a new performance experiment or an additional accuracy claim.

Verification record

Record field	Complete after each review
Resume ID / version	_____
Model and version	_____
Review date	_____
Prompt version	LLM Verification Framework v1.0
Predictions reviewed	_____
Verified exact / partial	_____ / _____
Unsupported / missed	_____ / _____
Human reviewer	_____

Final release checklist

[ ] Personal information was removed or handled with permission.

[ ] The original PDF and plain-text extraction were compared.

[ ] Every LLM decision was checked against resume evidence.

[ ] Unsupported qualifications were not added to the resume.

[ ] Model, date, prompt version, and human decisions were recorded.

[ ] The final PDF was exported and checked again after revision.

Final reminder: Use AI to locate possible problems faster. Use the original resume and human judgment to decide what is true.